

Introduction

Coordinating multiple robots in complex environments is a fundamental challenge in robotics. **Traditional sampling-based planners** such as RRT and PRM struggle with the **high dimensionality** and **computational overhead** of multi-robot settings. **Diffusion models** are generative algorithms that learn to denoise noise into target distributions. They produce **smooth, diverse, task-aware trajectories** and accept collision penalties as guidance during sampling.

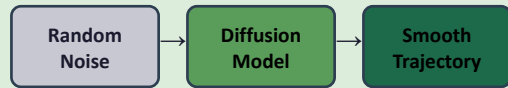


Fig. 1: Diffusion-based trajectory generation pipeline.

Recent work — **MPD** (Carvalho et al. 2024), **Projected Diffusion** (Liang et al. 2025), and **MMD** (Shaoul et al. 2025) — shows diffusion models can **scale to multi-robot planning**.

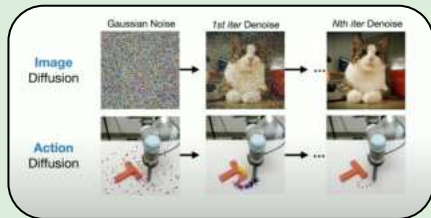


Figure 2. Comparison of image diffusion and action diffusion pipelines illustrating the denoising process from Gaussian noise to target output. Adapted from Yang et al. (2023).

Research Question

How can **diffusion models** be enhanced to generate **collision-free, efficient trajectories** in **complex multi-robot environments**?

Hypothesis: diffusion priors paired with classical safety enforcement will outperform RRT baselines in smoothness and efficiency.

References

- Carvalho, J., Le, A. T., Baieri, M., Koert, D., & Peters, J. (2024). Motion planning diffusion: Learning and adapting robot motion planning with diffusion models. *IEEE Transactions on Robotics*, 40, 1–18.
- Liang, J., Shaoul, Y., Kondapally, P., Patel, R., & Likhachev, M. (2025). Simultaneous multi-robot motion planning with projected diffusion models. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*. IEEE.
- Shaoul, Y., Liang, J., Kondapally, P., & Likhachev, M. (2025). Multi-robot motion planning with diffusion models. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE.
- Wang, J., Chi, W., Li, C., Wang, C., & Meng, M. Q.-H. (2021). Neural RRT*: Learning-based optimal path planning. *IEEE Transactions on Automation Science and Engineering*, 17(4), 1748–1758. <https://doi.org/10.1109/TASE.2020.2976560>
- Yang, L., Zhang, Z., Song, Y., Hong, S., Xu, R., Zhao, Y., Zhang, W., Cui, B., & Yang, M.-H. (2023). Diffusion models: A comprehensive survey of methods and applications. *ACM Computing Surveys*, 56(4), 1–39. <https://doi.org/10.1145/3626235>
- Zhang, Y. (2025). A survey of hybrid learning approaches for multi-robot coordination. *Journal of Field Robotics*, 42(2), 215–240.

Enhancing Multi-Robot Motion Planning with Diffusion Models for Multi-Objective Environments

Methods

- 1 Preparation**
Generated 12 simulated MRMP runs

5 robots per run in 10×10 environments

Randomized obstacles & start/goal positions

Trajectories saved as JSON
- 2 Planning**
RRT-based planner as baseline

Constructed configuration space

Filtered demonstrations for validity

Aligned to Liang et al. (2025) benchmark
- 3 Evaluation**
Computed path length & displacement
path efficiency = D / L

Smoothness = mean acceleration

Weighted composite accuracy score
- 4 Analysis**
Per-robot & per-run summary metrics

Efficiency–smoothness tradeoff plots

Path optimality scatter analysis

Aggregated across all 12 runs

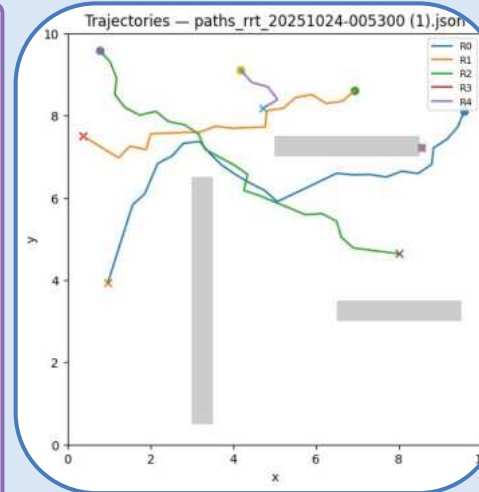


Figure 3. RRT-generated trajectories for five robots (R0–R4) navigating a 10×10 obstacle environment. Circles = start positions; X markers = goal positions.

Results

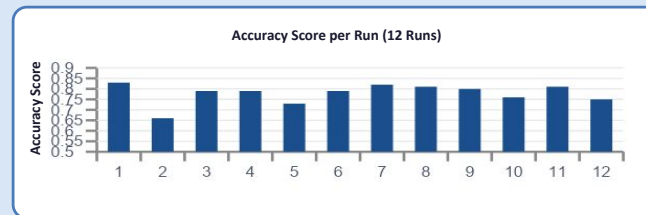
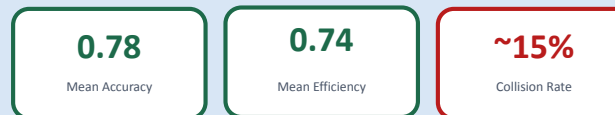


Figure 4. Mean composite accuracy score per run across 12 simulated RRT-based planning episodes. Higher values indicate better trajectory quality.

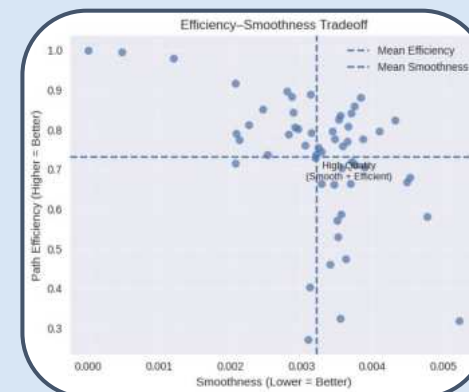


Figure 5. Efficiency–smoothness tradeoff across all valid robot trajectories. Dashed lines indicate mean efficiency and mean smoothness. Upper-left region = high-quality trajectories.

Conclusions

Do diffusion models hold promise for MRMP?

Yes. The baseline reveals a clear efficiency–smoothness tradeoff that diffusion priors are well-suited to address, since they can encode smooth, multi-modal trajectory distributions learned from demonstration data.

What does the baseline reveal?

Across 12 RRT runs, mean efficiency was only 0.74 — paths were ~35% longer than the straight-line minimum, and a residual 15% collision rate persisted across episodes. This headroom motivates the hybrid approach.

What is the hybrid paradigm?

Diffusion models propose smooth, task-aware trajectories, while classical search layers enforce hard safety constraints.

Future Research

- Train a diffusion model and compare directly to RRT baseline
- Add dynamic obstacles and real-world map variety
- Reduce collision rate to below 5% via constraint projection
- Scale to 40+ robot fleets with real-time replanning